

Tehman Hassan

Final Year Computer Engineering Student
Department of Computer Engineering, University of Engineering
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2023-UET/CP-011
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Profile and Bio

I am an independent and hardworking individual who enjoys continuous learning and working with a Team. My interests lie in Backend Systems, Artificial Intelligence and Scalable Software development using System Design.

Through my academic and independent work, I have gained experience in building REST APIs using Python Django REST Framework. In regards with ML and AI. My focus is to combine efficient System Design Concepts with the architectural view and then design software systems that combine algorithms with Practical solutions.

I have led 3 academic project teams in OOP, DBMS, and Data Structures courses, managing task allocation, mentoring teammates, and ensuring alignment with deadlines, documentation, and reports. I also held leadership roles in college, serving as an Appointment Holder responsible for discipline and extracurricular activities, and as Head of the Computing Club where I taught Python and HTML to junior students.

Education

Bachelor of Engineering in Computer Engineering

Aug 2023 - Present

[University of Engineering and Technology Taxila](#), Pakistan
Currently in 6th Semester, having an updated CGPA of 3.38

Matric (Primary) and High School Intermediate

Apr 2017- Oct 2022

[Cadet College Choa Saiden Shah Chakwal](#), Pakistan

A+ Grade. Activities and societies: Member of College Computing Club, Football Team, Debating Society, Appointment Holder
An institute that provided a disciplined and rigorous academic environment, focusing on character development and academic excellence. My time there involved a well-rounded curriculum and activities that shaped my foundation for further education.

Projects

Currently Leading Final Year Project with 2 fellow Students

Built "talentgraph.ai", talent mapping in Global AI Hackathon

Project Link: <https://skillpath.duckdns.org/>

A platform that allows workers especially tech and non-tech to map their skills on a global scale, allow a talented man in a small city to get a global view of his working niche, people and trends. Organization hiring can also be managed using AI based Description understanding and then showing talents across the globe in clusters good for this niche.

Full Stack Health Platform for Luther Health LTD

Project Link: <https://luther.health/>

Developed a Production Grade Health Management Platform for a Doctor in UK, Luther Health Limited using microservices based backend architecture. The system includes user authentication, patient health record management, appointment scheduling etc. Reduced **AI infrastructure deployment costs** by designing a workflow-based architecture using **n8n and AWS**.

Single Cycle Processor Datapath in Verilog HDL

Project Link: <https://github.com/Tehman700/Verilog-HDL>

Implemented a **Single-Cycle RISC-V Processor Datapath in Verilog HDL**, supporting multiple instruction types (R, I, Load, Store, Branch) and executing the full instruction cycle within one clock cycle. The processor architecture was built from scratch by implementing core hardware modules such as PC, IM, Register File, CU, ALU, ALU Control, and Memory.

Real-Time Asynchronous Chat App Using WebSocket's with End-to-End Encryption

Project Link: <https://github.com/Tehman700/Real-Time-Chat-App-2-Way-Communication-WebSockets>

This project is a full-stack real-time chat application that enables users to communicate instantly through a secure, asynchronous, two-way channel. The application is designed with modern security and usability features, including end-to-end encryption, OTP-based user verification, password reset with rate-limiting, and offline message handling.

Version Control System (Git-like) using Database

A fully custom version control system engineered from the ground up using database storage instead of a file-object structure. Supports commit creation, branching, restoring versions, file tracking, and repository history management. Inspired by Git internals and built as part of a deep dive into compilers, file systems, and backend system architecture.

Potholes Complaint System (Winning Project) in OOP Course Exhibition

The project was a Pothole Complaint System, developed as an Android app. Here's how it works: Users can lodge complaints by simply capturing a photo of a pothole. Azure Vision AI is used to determine whether the uploaded image is genuine or fake. Valid complaints are uploaded to a Spring Boot server hosted on Azure. Admin which currently we are can then manage these complaints and delete reports that have been resolved.

All data (JSON) is stored in a PostgreSQL database for efficient record-keeping. Since we studied Java for our OOP course, the project was primarily developed in Java.

AI Agent for Teachers (CLO & Bloom's Taxonomy Report Generator)

Project Link: https://github.com/Tehman700/Dr.Fawad_AI_Agent_Course

A smart Java-based AI agent that analyses academic documents and auto-generates structured reports aligned with CLOs and Bloom's Taxonomy levels. Uses OpenAI GPT models for classification and summaries. Features real-time updates, tree data structures for processing, vector ID storage for offline reliability, and seamless auto PDF generation.

Working on currently to shift this whole architecture on n8n and hosted on Cloud as AWS with Gemini API for Minimal COST.

Work Experience

Freelance Backend Software Developer Python

Aug 2025 – Jan 2026

Fiverr, Freelance Platform

- Main tech stack majorly being used is Python, Django, REST APIs, PostgreSQL, Django Channels, WebSocket's, creating CI/CD pipelines for automated deployments on AWS cloud services, subscriptions, and activity logging using signals in the Framework.
- Reached Level 1 Seller in less than 2 months, Error handling, troubleshooting on different platforms including AWS, Nginx servers.

Software Engineering Backend Intern

Jul 2025 – Sep 2025

Inara Technologies, F11 Markaz, Islamabad, Pakistan

- Developed a full-stack REST API from scratch that handles all Custom Exceptions, JWT authentication, CRUD, Nested Comments and replies, likes, dislikes, for every module, role-based permissions, subscriptions, and activity logging using signals in the Framework.
- Further worked on WebSocket's Concepts for Real-Time two-way communication from both ends. GitHub Actions and AWS Pipelines as also learnt along the way.

Competitions

Winner OOP Course Semester Project

Won the best Semester Project in my department among 150 students, along with certificates and prize money

Participation in Hackathon in Devsinc

This was a Speed Programming Competition in a Professional Software Company, tackling Data Structures, Greedy Algorithms, and DP Problems

Certificates

- API Testing Learning Path Certificate
- Postman API Fundamentals Student Expert
- Machine Learning with Python
- 100 Days of Code: The Complete Python Pro Bootcamp
- SQL Basic Hacker rank
- Transformers in LLMs Critical
- Thinking and Problem Solving (LinkedIn)
- Participation in Space Week Organized by NCGSA